

Lecture 7

Query Algorithms and Phase Kick-back

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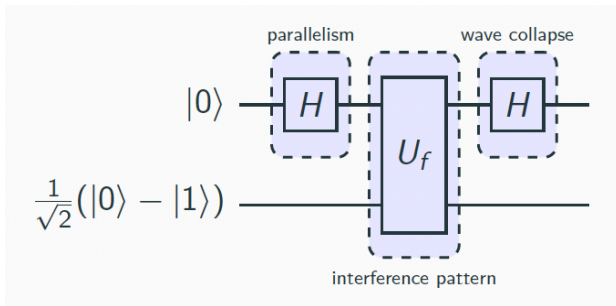
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Revisiting the Deutsch algorithm



Basic ingredients

- Input in **superposition** (leading to some form of 'natural parallelism')
- An **oracle** for f taking the form of a **controlled gate** on the input
- A specific **preparation of the first qubit** (to induce some sort of interaction)

Input on demand

Usually, input is provided to the computational process once and for all in the form of a **value**, e.g., a binary string.

In this class of algorithms, however, the input is **hidden**, provided as a **function**

$$f : 2^n \longrightarrow 2^m$$

to which the process has limited access: it can only evaluate it on specific values.

A query selects a specific string $s \in 2^n$ and the **oracle** returns to the computational process the string $f(s) \in 2^m$.

The **efficiency** of an algorithm in this model is related to the **number of queries** it requires.

Input on demand

Examples of query problems

- **Input:** $f : 2^n \longrightarrow 2$
Problem: Decide if there is a string s st $f(s) = 1$.
- **Input:** $f : 2^n \longrightarrow 2^m$
Problem: Output the smallest string in the lexicographic order on 2^m (assuming the function represents integers, the problem is to find their minimum)
- **Input:** $f : 2^n \longrightarrow 2$ with a **promise** (e.g. there is only one string s for which $f(s) = 1$).
Problem: Find s

Oracle-based computation
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Phase kick-back
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Bernstein-Vazirani's problem
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Deutsch-Josza's problem
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as discussed in the previous lecture.

Example

Let $f : \{0, 1\}^2 \rightarrow \{0, 1\}$ be such that $f(01) = 1$ and evaluates to 0 otherwise.

Oracle $U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle$ can be tabulated as

$$\begin{array}{ll} |00\rangle|0\rangle \mapsto |00\rangle|0\rangle & |00\rangle|1\rangle \mapsto |00\rangle|1\rangle \\ |01\rangle|0\rangle \mapsto |01\rangle|1\rangle & |01\rangle|1\rangle \mapsto |01\rangle|0\rangle \\ |10\rangle|0\rangle \mapsto |10\rangle|0\rangle & |10\rangle|1\rangle \mapsto |10\rangle|1\rangle \\ |11\rangle|0\rangle \mapsto |11\rangle|0\rangle & |11\rangle|1\rangle \mapsto |11\rangle|1\rangle \end{array}$$

which corresponds to

$$\begin{aligned} \sum_{x \in \{0,1\}^2} |x\rangle\langle x| X^{f(x)} &= \\ &= |00\rangle\langle 00| \otimes I + |01\rangle\langle 01| \otimes X + |10\rangle\langle 10| \otimes I + |11\rangle\langle 11| \otimes I \end{aligned}$$

Example

Or, in matrix format,

$$U_f = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & X & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Back to the Deutsch algorithm

What is new in quantum evaluation of Boolean functions is the ability to act on a **superposition**, e.g.

$$\sum_x |x\rangle|0\rangle \mapsto \sum_x |x\rangle|f(x)\rangle$$

i.e. **all results are computed in a single execution**

But also interesting is the effect of **starting with $|-\rangle$** :

$$\sum_x |x\rangle|-\rangle \mapsto \sum_x (-1)^{f(x)} |x\rangle|-\rangle$$

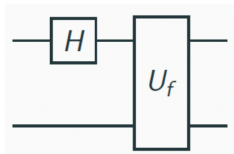
which indeed generates the **suitable** interference.

Revisiting the Deutsch algorithm

The oracle

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$$

leaves the first qubit unchanged, but modifies the second one.
Look again, however, how the oracle behaves in the circuit



behaves when processing $H|0\rangle|-\rangle = |+\rangle|-\rangle$, leading to

$$\frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle|-\rangle + (-1)^{f(1)} |1\rangle|-\rangle \right)$$

Pulling $(-1)^{f(0)}$ out of the sum, yields

$$\frac{1}{\sqrt{2}} (-1)^{f(0)} \left(|0\rangle|-\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle|-\rangle \right)$$

Revisiting the Deutsch algorithm

Making the conditional structure explicit, this equivaless to

$$\begin{cases} (\pm 1)|+\rangle|-\rangle & \text{if } f(0) \oplus f(1) = 0 \text{ (i.e., } f(0) = f(1)) \\ (\pm 1)|-\rangle|-\rangle & \text{if } f(0) \oplus f(1) = 1 \text{ (i.e., } f(0) \neq f(1)) \end{cases}$$

which later, after applying the final H to the first qubit, leads to

$$\begin{cases} (\pm 1)|0\rangle|-\rangle & \text{if } f(0) = f(1) \\ (\pm 1)|1\rangle|-\rangle & \text{if } f(0) \neq f(1) \end{cases}$$

providing a solution to the Deutsch problem.

But now the second qubit remains unchanged and a phase factor is **kicked back** to the first one.

What's under the carpet?

Let's try to understand what explains this **kick back** behaviour. Clearly,

$$|b \oplus 1\rangle = |\neg b\rangle = X|b\rangle = X^1|b\rangle$$

$$|b \oplus 0\rangle = |b\rangle = I|b\rangle = X^0|b\rangle$$

that is,

$$|b \oplus c\rangle = X^c|b\rangle \text{ for } b, c \in 2$$

Therefore,

$$U_f(|a\rangle|b\rangle) = |a\rangle|b \oplus f(a)\rangle = |a\rangle X^{f(a)}|b\rangle \text{ for } a, b \in 2$$

and, by **linearity** (why?),

$$U_f(|a\rangle|v\rangle) = |a\rangle X^{f(a)}|v\rangle \text{ for any vector } |v\rangle$$

What's under the carpet?

Making $|v\rangle = |-\rangle$, yields

$$U_f(|a\rangle|-\rangle) = |a\rangle X^{f(a)}|-\rangle$$

Finally

$$U_f(|a\rangle|-\rangle) = \begin{cases} |a\rangle I|-\rangle = |a\rangle|-\rangle & \text{if } f(a) = 0 \\ |a\rangle X|-\rangle = |a\rangle(-|-\rangle) = -|a\rangle|-\rangle & \text{if } f(a) = 1 \end{cases}$$

because

$$I|-\rangle = |-\rangle \text{ and } X|-\rangle = -|-\rangle$$

which means that $(-1, |-\rangle)$ is a **eigen value/vector** pair for operator X (and, similarly, $(1, |-\rangle)$ for I).

A even simpler example: the behaviour of cX



$$\overbrace{\begin{bmatrix} I & 0 \\ 0 & X \end{bmatrix}}^{cX}$$

i.e., $|xy\rangle \mapsto |x, x \oplus y\rangle$

where $|x\rangle$ is called the **control** qubit, and $|y\rangle$ the **target** qubit.
Clearly,

$$cX|0\rangle|\varphi\rangle = |0\rangle I|\varphi\rangle$$

$$cX|1\rangle|\varphi\rangle = |1\rangle X|\varphi\rangle$$

Seen as an oracle, note that **input** is presented at the **control** qubit and **output** is produced on the **target** qubit.

A even simpler example: the behaviour of cX

Consider now a special case: prepare the target qubit with $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ which, as seen before, is an **eigenvector** of both

- X (with $\lambda = -1$) and of I (with $\lambda = 1$)
- and, thus, $X \frac{|0\rangle - |1\rangle}{\sqrt{2}} = -1 \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ and $I \frac{|0\rangle - |1\rangle}{\sqrt{2}} = 1 \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

Therefore,

$$\begin{aligned} cX|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= |1\rangle \left(X \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) \\ &= |1\rangle \left((-1) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) \\ &= -|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

$$\text{while } cX|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

A even simpler example: the behaviour of cX

A phase (1 or -1 , i.e., the corresponding **eigenvalue**)

jumps, or is **kicked back**

from the **second** (target) to the **first** (control) qubit where the input is presented.

This effect is suitably recorded in the following formulation of cX :

$$cX|b\rangle|-\rangle = (-1)^b|b\rangle|-\rangle \quad \text{with } b \in \mathbf{2}$$

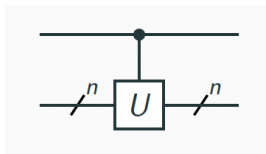
Observe now that, through the kick-back effect, **ouput** arises in the **control** qubit, whereas the **target** qubit remains unchanged.

Example:

$$cX \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

The phase kick-back pattern

This can be generalised to every **controlled** quantum operation:



Let v be an eigenvector of U with eigenvalue $e^{i\theta}$, (i.e. $Uv = e^{i\theta}v$). Thus,

$$\begin{aligned} & cU((\alpha|0\rangle + \beta|1\rangle) \otimes v) \\ &= cU(\alpha|0\rangle \otimes v + \beta|1\rangle \otimes v) \\ &= \alpha|0\rangle \otimes v + \beta|1\rangle \otimes Uv \\ &= \alpha|0\rangle \otimes v + \beta|1\rangle \otimes e^{i\theta}v \\ &= \alpha|0\rangle \otimes v + e^{i\theta}\beta|1\rangle \otimes v \\ &= (\alpha|0\rangle + e^{i\theta}\beta|1\rangle) \otimes v \end{aligned}$$

The phase kick-back pattern

- Global phase $e^{i\theta}$ (introduced to v) was 'kicked-back' as a relative phase in the control qubit
- Some information of U is now encoded in the control qubit

In general kicking-back such phases causes interference patterns that give away information about U .

Our two examples

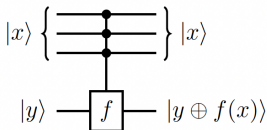
in the cX gate: $cX|b\rangle|-\rangle = (-1)^b|b\rangle|-\rangle$

in Deutsch algorithm: $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$

Note

The oracle in the Deutsch algorithm takes the form of a **generalised cX** :

$$\sum_{x \in \{0,1\}^n} |x\rangle\langle x| X^{f(x)}$$



The Bernstein-Vazirani algorithm

Let $2^n = \{0, 1\}^n = \{0, 1, 2, \dots, 2^n - 1\}$ be the set of non-negative integers (represented as bit strings up to n bits).

The problem

Let s be an unknown non-negative integer less than 2^n , encoded as a bit string, and consider a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ which hides secret s as follows: $f(x) = x \cdot s$, where $\cdot$ is the bitwise product of x and s modulo 2. i.e.

$$x \cdot s = x_1 s_1 \oplus x_2 s_2 \oplus \dots \oplus x_n s_n$$

Find s .

Note that juxtaposition abbreviates conjunction, i.e., $x_1 s_1 = x_1 \wedge s_1$

Setting the stage

Lemma

(1) For $a, b \in 2$ the equation $(-1)^a(-1)^b = (-1)^{a \oplus b}$ holds.

Proof sketch

Build a truth table for each case and compare the corresponding contents.

Lemma

(2) For any three binary strings $x, a, b \in 2^n$ the equation $(x \cdot a) \oplus (x \cdot b) = x \cdot (a \oplus b)$ holds.

Proof sketch

Follows from the fact that for any three bits $a, b, c \in 2$ the equation $(a \wedge b) \oplus (a \wedge c) = a \wedge (b \oplus c)$ holds.

Setting the stage

Lemma

(3) For any element $|b\rangle$ in the computational basis of $\mathbb{C}^2$,

$$H|b\rangle = \frac{1}{\sqrt{2}} \sum_{z \in 2} (-1)^{b \cdot z} |z\rangle$$

Proof sketch

Build a truth table and compare the corresponding contents.

Theorem

(1) For any element $|b\rangle$ in the computational basis of $\mathbb{C}^{2^n}$,

$$H^{\otimes n}|b\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in 2^n} (-1)^{b \cdot z} |z\rangle$$

Proof sketch

Follows by induction on the size of n .

The Bernstein-Vazirani algorithm

How many times f has to be called to determine s ?

- Classically, we run f n -times by computing

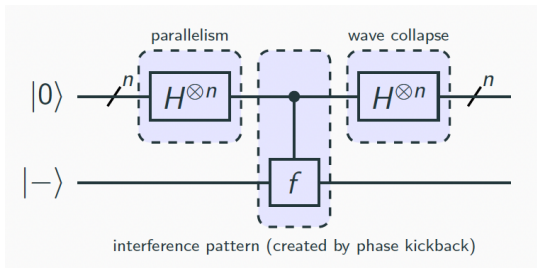
$$f(1 \dots 0) = (s_1 \wedge 1) \oplus \dots \oplus (s_n \wedge 0) = s_1$$

$$\vdots$$

$$f(0 \dots 1) = (s_1 \wedge 0) \oplus \dots \oplus (s_n \wedge 1) = s_n$$

- With a quantum algorithm, we may discover s by running f only once

The circuit



The computation

$$\begin{aligned}
 & H^{\otimes n} |0\rangle |-\rangle \\
 &= \frac{1}{\sqrt{2^n}} \sum_{z \in 2^n} |z\rangle |-\rangle && \{\text{Theorem 1}\} \\
 &\xrightarrow{U_f} \frac{1}{\sqrt{2^n}} \sum_{z \in 2^n} (-1)^{f(z)} |z\rangle |-\rangle && \{\text{Defn. } U_f\} \\
 &\xrightarrow{H^{\otimes n} \otimes I} \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{f(z)} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) |-\rangle && \{\text{Theorem 1}\} \\
 &= \frac{1}{2^n} \sum_{z \in 2^n} \sum_{z' \in 2^n} (-1)^{(z \cdot s) \oplus (z \cdot z')} |z'\rangle |-\rangle && \{\text{Lemma 1 and defn. } f\} \\
 &= \frac{1}{2^n} \sum_{z \in 2^n} \sum_{z' \in 2^n} (-1)^{z \cdot (s \oplus z')} |z'\rangle |-\rangle && \{\text{Lemma 2}\} \\
 &= |s\rangle |-\rangle && \{\text{Why?}\}
 \end{aligned}$$

Why?

$$\dots = \frac{1}{2^n} \sum_{z \in 2^n} \sum_{z' \in 2^n} (-1)^{z \cdot (s \oplus z')} |z'\rangle |-\rangle = \dots$$

For each binary string $s \oplus z'$, $\frac{1}{2^n} \sum_{z=0}^{2^n-1} (-1)^{z \cdot (s \oplus z')}$ is **1** iff $(s \oplus z') = 0$, which happens only if $s = z'$.

In all other cases $\frac{1}{2^n} \sum_{z=0}^{2^n-1} (-1)^{z \cdot (s \oplus z')}$ is **0**.

Thus, the only non-zero amplitude is the one associated with s .

Why?

The conclusion comes from the following result: for each binary string x ,

$$\frac{1}{2^n} \sum_{z=0}^{2^n-1} (-1)^{z \cdot x} = \begin{cases} 1 & \text{if } x = 0^n \\ 0 & \text{if } x \neq 0^n \end{cases}$$

Proof

- for $x = 0$, $\frac{1}{2^n} \sum_{z=0}^{2^n-1} (-1)^{z \cdot x} = \frac{1}{2^n} \sum_{z=0}^{2^n-1} 1 = 1$.
- for $x \neq 0$, half of the 2^n factors in the sum will be -1 and the other half 1 , thus summing up to 0 . This is because the value of $z \cdot x$ is 1 for an odd number of coincidences of 1 bits in both arguments, and 0 otherwise, and z spans all numbers from 0 to $2^n - 1$

Why?

Alternatively, consider the probability of measuring s at the end of the computation:

$$\begin{aligned}
 & \left| \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{z \cdot (s \oplus s)} \right|^2 \\
 &= \left| \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{z \cdot 0} \right|^2 \\
 &= \left| \frac{1}{2^n} \sum_{z \in 2^n} 1 \right|^2 \\
 &= \left| \frac{2^n}{2^n} \right|^2 \\
 &= 1
 \end{aligned}$$

This means that somehow all values yielding wrong answers were completely **cancelled**.

Deutsch-Josza

The Problem

Take a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, which is known to be either constant or balanced.

Find out which case holds.

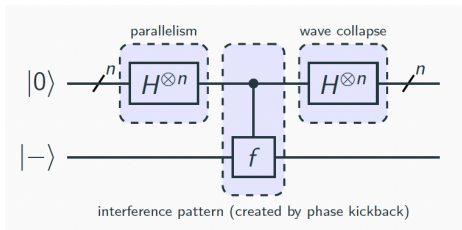
Classically, we evaluate half of the inputs ($\frac{2^n}{2} = 2^{n-1}$), evaluate one more and run the decision procedure,

- output always the same $\implies$ constant
- otherwise $\implies$ balanced

which requires running f $2^{n-1} + 1$ times.

A quantum algorithm replies by running f only once.

The circuit



Actually, the algorithm is the **same** used to solve the Bernstein-Varziani problem.

More precisely, the problem is solved with the Bernstein-Varziani algorithm **for a function satisfying the Deutsch-Joza promise**.

The computation

$$\begin{aligned}
 & H^{\otimes n} |0\rangle |-\rangle \\
 &= \frac{1}{\sqrt{2^n}} \sum_{z \in 2^n} |z\rangle |-\rangle && \{\text{Theorem 1}\} \\
 &\xrightarrow{U_f} \frac{1}{\sqrt{2^n}} \sum_{z \in 2^n} (-1)^{f(z)} |z\rangle |-\rangle && \{\text{Defn. } U_f\} \\
 &\xrightarrow{H^{\otimes n} \otimes I} \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{f(z)} \underbrace{\left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right)}_{\square \text{ upper qubits}} |-\rangle && \{\text{Theorem 1}\}
 \end{aligned}$$

Developing $\square$ by case distinction

f is constant

$$\begin{aligned} & \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{f(z)} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \\ &= \frac{1}{2^n} (\pm 1) \sum_{z \in 2^n} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \end{aligned}$$

Therefore, at state $|0\rangle$ is

$$\boxed{f \text{ is constant at } 1} \rightsquigarrow \frac{-(2^n)}{2^n} |0\rangle = -|0\rangle$$

$$\boxed{f \text{ is constant at } 0} \rightsquigarrow \frac{(2^n)}{2^n} |0\rangle = |0\rangle$$

Developing $\square$ by case distinction

Actually the probability of measuring $|0\rangle$ at the end given by

$$\begin{aligned} & \left| \frac{1}{2^n} (\pm 1) \sum_{z \in 2^n} (-1)^{z \cdot 0} \right|^2 \\ &= \left| \frac{1}{2^n} (\pm 1) \sum_{z \in 2^n} 1 \right|^2 \\ &= \left| \frac{2^n}{2^n} \right|^2 \\ &= 1 \end{aligned}$$

So if f is constant we measure $|0\rangle$ with probability 1.

Developing $\square$ by case distinction

f is balanced

$$\begin{aligned}
 & \frac{1}{2^n} \sum_{z \in 2^n} (-1)^{f(z)} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \\
 &= \frac{1}{2^n} \left(\sum_{z \in 2^n, f(z)=0} (-1)^{f(z)} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \right. \\
 & \quad \left. + \sum_{z \in 2^n, f(z)=1} (-1)^{f(z)} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \right) \\
 &= \frac{1}{2^n} \left(\sum_{z \in 2^n, f(z)=0} \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \right. \\
 & \quad \left. + \sum_{z \in 2^n, f(z)=1} (-1) \left(\sum_{z' \in 2^n} (-1)^{z \cdot z'} |z'\rangle \right) \right)
 \end{aligned}$$

Developing □ by case distinction

Probability of measuring $|0\rangle$ at the end given by

$$\begin{aligned} & \left| \frac{1}{2^n} \left(\sum_{z \in 2^n, f(z)=0} (-1)^{z \cdot 0} + \sum_{z \in 2^n, f(z)=1} (-1)(-1)^{z \cdot 0} \right) \right|^2 \\ &= \left| \frac{1}{2^n} \left(\sum_{z \in 2^n, f(z)=0} 1 + \sum_{z \in 2^n, f(z)=1} (-1) \right) \right|^2 \\ &= \left| \frac{1}{2^n} \left(\sum_{z \in 2^n, f(z)=0} 1 - \sum_{z \in 2^n, f(z)=1} 1 \right) \right|^2 \\ &= 0 \end{aligned}$$

So if f is balanced we measure $|0\rangle$ with probability 0

